



Society of Petroleum Engineers

SPE-230092-MS

Pioneering AI Driven Advanced Process Control: Transforming Upstream Oil & Gas Operational Landscape

Usama Abdul Zahir, Salma Almazrouei, Awais Ashraf, Jigar Modi, Syed Ajmal, and Gaurav Gupta, ADNOC Onshore

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/230092-MS](https://doi.org/10.2118/230092-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

This paper narrates the successful deployment of AI-enabled Advanced Process Control (APC) to enhance operational efficiency, automate operator actions, and optimize energy consumption in oil and gas upstream processing facilities. The approach utilizes data-driven machine learning models to predict GOSP (Gas Oil Separation Plants) train performance parameters to achieve following objectives:

- Maximize oil production.
- Maintain crude specifications.
- Optimize energy and wash water consumption.
- Slug detection and disturbance rejection.

Introduction

This paper narrates the successful deployment of AI-enabled Advanced Process Control (APC) to enhance operational efficiency, automate operator actions, and optimize energy consumption in oil and gas upstream processing facilities. The approach utilizes data-driven machine learning models to predict GOSP (Gas Oil Separation Plants) train performance parameters to achieve following objectives:

- Maximize oil production.
- Maintain crude specifications.
- Optimize energy and wash water consumption.
- Slug detection and disturbance rejection.

Process Description

The process flow diagram (PFD) of central processing Facility (CPF) of reference upstream oil and gas field shown in [Fig.1](#), is designed to treat well fluid from the new and re-routed wells. The overall capacity is

approximately 140 MBPOD. The multiphase fluid is routed to the CPP via two trunklines TL-3 and TL-4, where it is separated into oil, gas and water. Each of these fluids will undergo further treatments and are routed to the relevant receiving facilities.

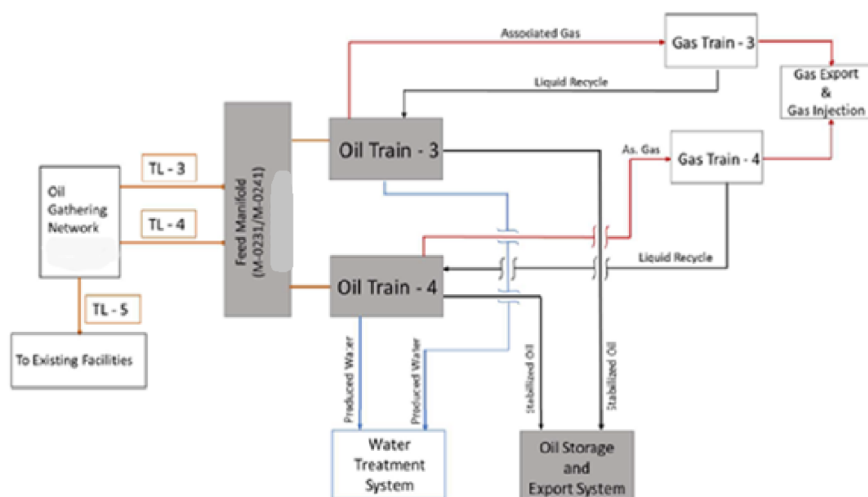


Figure 1—Upstream Central Processing Plant Process Flow Diagram (PFD)

Oil is processed in two identical trains, each designed to treat 70 MBPOD. Each train consists of a primary and a secondary separation stage, a dehydration and desalting stage and finally a stabilization stage before being sent to the oil storage and export facilities. Associated gas is sent to the gas processing trains for conditioning, compression and re-injection. Produced gas capacity is higher than the capacity needed to be re-injected into the WAG wells. The annual gas surplus is sent to export facilities, compressed, combined with nearby field gas, and exported. Produced water is sent to the Produced Water Treatment Plant for oil removal and sub-surface disposal by re-injection to Water Disposal Wells. Following is a brief of major equipment in the GOSP trains a.

- **SLUG CATCHER**

The two slug catchers are identically designed for two-phase separation (gas and bulk liquid) and can each hold 582 m³ of slug volume to handle the largest surge from either trunkline during pigging. Separated gas and liquid are sent to the first stage separator. Gas and liquid from trains 3 and 4 share a common header before splitting into two identical lines to ensure an even 50% distribution between the trains.

- **1st STAGE SEPARATOR**

Liquid from each slug catcher is combined into a common manifold, then split into two sections to supply the separator trains. The first-stage, three-phase separator has compartments for oil and water; gas goes to MP compression, oil to the crude heater's tube side, and water to the Produced Water Treatment system.

- **CRUDE HEATER**

Liquids from LP compressor suction and interstage scrubber, MP compressor suction scrubber, closed drain pumps, and oil recovery drum pump are combined with crude from the first stage separator and sent to the crude heater. Each train has a 2 × 50% crude heater configuration. Oil from the first stage separator is preheated in the crude heater tubes to 60 °C (winter) or 75 °C (summer) before entering the second stage separator. This preheating boost vapor recovery in the second stage separator. Additionally, hot stabilized oil from the stabilizer column bottom preheats the incoming oil through a feed/product heat exchanger.

- **2nd STAGE SEPARATOR**

Preheated oil from the crude heater is flashed in the three-phase 2nd stage separator. Separator gas goes to LP compression, oil is pumped to the desalter package, and water is sent to the Produced Water Treatment system.

- **DESALTER PACKAGE**

Oil from the 2nd Stage Separator is pumped to the Desalters/Dehydrators package, where feed pumps increase its pressure from 5 to 11.6 barg. The oil undergoes dehydration and salt removal via electrostatic coalescence in two stages, meeting specifications of 0.05% BS&W and 10 PTB salt. The treated oil is then sent to the stabilization column for further processing.

- **OIL STABILIZATION**

Oil from the 2nd Stage Desalter goes to the Oil Stabilization Column, where lighter ends are stripped to meet TVP (10 psia $\pm$ 0.2 in winter) and H₂S (35 ppmw) specs, and any remaining water is removed. Oil enters at the column top, and liquid taken just below tray 19 is sent to the oily water separator—water then goes to produced water treatment, and oil is recycled back to the column. Wet sour gas from the stabilization column is routed to LP compression trains along with associated gas, all under pressure control. Liquid from column chimney tray (Tray 1) is sent to two Oil Stabilization Column Reboilers (one operating, one standby), where it is heated to about 180°C by a 220°C heating medium. The heated oil and vapour return to the stabilization column. After heat recovery, stabilized oil is directed to flow suction tanks.

- **HOT OIL SYSTEM**

Hot oil serves as the heating medium for stabilizer and glycol regeneration column re-boilers. The system includes circulation pumps, a heater, an expansion vessel (operating at 4–4.5 barg), a sump drum, an unloading package, and a trim cooler. Hot oil exits the heater at 220°C and enters the expansion vessel at 194–198°C.

Problem Statement / Challenge

The evaluation of plant performance indicators has revealed several key challenges that can be mitigated through the adoption of advanced technologies aimed at improving operational efficiency and excellence, as outlined below:

- A reactive approach is employed to maintain crude specifications such as RVP/TVP, H₂S levels, and salt content, due to the absence of online analyzers for real-time monitoring. Operators rely on daily laboratory sampling and test results to make process adjustments necessary to achieve target crude specifications.
- This methodology necessitates frequent operator involvement, thereby increasing personnel workload.
- Elevated standard deviations observed in process flows, pressures, and levels contribute to facility trips during operational disturbances.
- There are further opportunities to enhance the efficiency of utility consumption, particularly regarding freshwater and energy use.
- There is significant dependence on the experience and competence of Control Room Operators (CROs) to ensure the safe operation and control of the facility to meet production mandates.

The Solution

An Ai-Enabled Adaptive Process Control technology based Advanced Process Control (APC) was carefully selected to co-develop and deploy a robust APC solution in closed loop that can deliver following minimum requirements:

- Employs techniques that reduce model-building complexity and build time.
- Utilizes embedded AI algorithms & Deep Learning techniques.
- Execute parallel computing to build and deploy seed models online as self-learning.
- Model performance and prediction accuracy of more than 90% to execute energy efficient control actions.
- Sustain APC performance by auto-updating models online with nondisruptive background testing and no requirement for cloud connection.
- Monitor and record APC system Performance (such as percentage of time APC controller is ON, MV Utilization factor, percentage of CVs in service or constraint violation, percentage of MVs in control, etc.)

APC Objectives & SMART KPIs

Workshops were conducted with operations, engineering and technology teams and primary objectives of advanced process control (APC) were developed focusing on process performance optimization, efficiency improvement, and automating operator's actions pertaining to smooth and stable operations. Following SMART KPIs were set as a guiding principle to design customized APC model and for post-deployment performance evaluation:

- Maximize oil production (1-2%)
- Maintain TVP (True Vapor Pressure) of the stabilized crude within 10 psi +/- 0.2 psi for each train along with H₂S content limit to less than 35 ppm
- Optimize energy consumption (5-15%)
- Slug Detection and Disturbance Rejection (5-10%)
- Improve column stabilization (5-10%)
- Wash water optimization (1-3%)
- Reduce the Operator interventions (up to 25%)

Multiple simulations were performed to set the target values of above-mentioned KPIS. Duration of Evaluation was set be One month after deployment of APC Model (with APC Online time greater than minimum threshold).

Methodology & Project Execution Strategy

Initially, plant data was analyzed to develop dynamic linear models of the GOSP trains and design inferential quality soft sensors for real-time monitoring and control of plant operations. Subsequently, these models formed the basis for APC strategies aimed at enhancing oil production, optimizing energy consumption, and stabilizing facility operations without human intervention. Additionally, the deployed APC utilized a self-calibration mode that performed automatic background step testing while a robust algorithm ensured variables remained within operating limits, thereby eliminating the need for manual model tuning. The APC controller was validated through both open and closed loop simulations to ensure reliable performance.

A comprehensive strategy was established to formalize the end-to-end process of designing and collaboratively developing the APC solution within a closed-loop framework to fulfil the defined objectives. This strategy encompasses the following phases:

- Data Collection

- Site Survey
- Data Analysis & Baseline
- APC Application Design & Functional Design Specifications
- Gain Matrix for the Implemented APC controllers
- Implementation / System Design – DCS Interface
- Documentation / Training / Management of Change
- Post commissioning Performance Evaluation

Data Collection

The data collection process involved a combination of site surveys, review of plant process data, and analysis of engineering documentation. Process, operations, and engineering data for both Oil Trains were gathered as outlined below:

- Process description / Operating manual
- PFDs
- P&IDs
- DCS Graphics Screenshots
- DCS tag dumps include tag name, tag description, Unit of measurement, instrument range (low and high values), Tuning values, Filter, Alarm range, algorithm, etc.
- Process data for from Historian – two sets – one set for summertime – another set for wintertime
- Lab values of all sample points – two sets – one set of data for summertime – another set for wintertime
- ICSS Architecture

Data Analysis & Baseline. To evaluate the APC controller benefits and performance, 3 months each of summer and winter period uncompressed raw data was collected from historian and selected baseline period for the project is shown in [Table-1](#)

Table 1—Selected Baseline Period for APC Performance Evaluation Reference

Period	Start Date & Time	End Date & Time
Baseline Period (Winter)	1st Nov 20XX 00:05 hrs	22nd Jan 20XX 11:00 hrs
Baseline Period (Summer)	1st Jun 20XX 00:00 hrs	31st Aug 20XX 00:00 hrs

The Good period to be used for baseline analysis is identified by following tags with conditions as shown below.

Table 2—Baseline Data Conditioning

S. No	Tag Name	Description	UOM	Condition	Remarks
1.	0201FIC0231_02.PV	Train-3 Slug catcher oil flow	BPD	>15,000 BPD	Good data, can be used in baseline
2.	0201FIC0241_02.PV	Train-4 Slug catcher oil flow	BPD	>15,000 BPD	

S. No	Tag Name	Description	UOM	Condition	Remarks
3.	0201FIC6000_21.PV	Hot Oil Heater- HO Circulation flow	BPD	>300,000 BPD	

In addition to the conditions described above, instances of brief irregularities—such as sudden spikes or drops in data, and periods lacking data due to temporary connection issues—were excluded from the analysis, as illustrated in [Picture-1](#) and [Picture-2](#), which displays screenshots of data selection for both winter and summer periods respectively.

Source	Type	Start Time	End Time	Description
Dataset	BadSlice	11/17/2023 09:46:16	11/17/2023 10:50:59	
Dataset	BadSlice	12/30/2023 05:07:01	12/30/2023 23:15:02	No data
Dataset	BadSlice	1/18/2024 13:23:42	1/18/2024 22:11:03	Plant upset

Picture-1—Winter Period Bad Data Slices

Source	Type	Start Time	End Time	Description
Dataset	BadSlice	7/09/2023 04:20:05	7/09/2023 19:54:10	
Dataset	BadSlice	8/24/2023 01:45:42	8/24/2023 21:59:02	

Picture-2—Summer Period Bad Data Slices

This slicing operation was to remove bad data which may hamper the fair comparison between before and after implementation. Now that the dataset is fixed, post APC implementation data shall be compared with this final selected baseline period and target KPI values as listed in [Table-3](#).

Table 3—Statistical Values for Baseline (Time Weighted Average)

S No	APC Objective	KP	UOM	Statistical values for Baseline (Time weighted avg.)				KP Calculation Formula
				Min	Max	Avg	Std Dev.	
1	Maximize oil production (1-2%)	Recovered oil in T-3 Stabilizer	%	34.83	100	94.2	5.746	Average of Oil recovered % = (AVG(Baseline) – AVG(B/ aluationPeriod)) / AVG(Baseline) × 100
		Recovered oil in T-4 Stabilizer	%	65.2	100	94.4	3.083	
2	Maintain TVP (True Vapor Pressure) of the stabilized crude within 10 psi +/- 0.2 psi.	Crude Oil -TVP (Lab quality)	PS	6.8	8.8	7.771	0.578	Average of TVP quality = (AVG(Baseline) – AVG(B/ aluationPeriod)) / AVG(Baseline) × 100
	Energy Savings	Optimization of Lean gas consumption	Sm3/hr/1000 bbl of oil	12.8	210.8	24.49	5.821	Average of FG consumed = (Avg(Baseline) – Avg (evaluation)/ Avg(Baseline)) × 100
3	Minimize disturbances in Slug catcher	Variation in T-3 Slug catcher level	%	16.83	57.5	37.02	4.44	Improvement = (STDev(Baseline) – SI Dev(B/ aluationPeriod)) / SIDev(Baseline) × 100
		Variation in T-3 Slug catcher flow	BPD	15200	111200	68410	9782	

S No	APC Objective	KP	UOM	Statistical values for Baseline (Time weighted avg.)				KP Calculation Formula
				Min	Max	Avg	Std Dev.	
		Variation in T-4 Slug catcher level	%	21.22	59.16	37.56	4.035	
		Variation in T-4 Slug catcher flow	BPD	15000	128900	69640	14220	
4	Improve Column stabilization	Variation in T-3 Stab column temperature	Deg C	142.5	182.5	163.1	4.383	
		Variation in T-4 Stab column temperature	Deg C	129.7	186.5	162.7	3.8	
5	Improve Wash water consumption	T-3 Wash water to Oil ratio	BPD/bbl of oil	0.903	3.356	1.807	0.3513	Improvement = ((AVG(Baseline) – AVG(EvaluationPeriod)) / AVG(Baseline) × 100
		T-4 Wash water to Oil ratio	BPD/bbl of oil	1.586	3.448	1.737	0.3842	
6	Maintain H2S content in Stabilized crude	H2S Inferential	ppm	2	3.28	2.661	0.478	Improvement = (STDev(Baseline) – STDev(BvaluationPeriod)) / STDev(Baseline) × 100
7	Maintain Salt content in Stabilized crude	Salt Inferential	PTB	5	15	7.341	1.598	

Site Survey. Following completion of the data analysis and establishment of the baseline, a comprehensive site survey was undertaken to address the following checklists:

- Process operation clarifications
- APC Scope related Instruments status check
- APC Scope related Control valve status check
- PID tuning requirements for necessary loops
- Mode change test for potential APC MVs
- Step size determination for identified APC MVs
- Lab sampling requirements

A series of checklists were executed to establish the health and integrity of control loops in the scope of APC as listed in [Table-4](#).

Table 4—APC Scope Control Loops Health Status

S.No	Section	Instrument Tag name	Description	Instrument Status
1	Train-3 (Slug catcher)	0201PI1391_04.PV	Trunkline-3 pressure	GOOD
2		0201PI1392_04.PV	Trunkline-4 pressure	GOOD
3		0201FIC0231_02.PV	Slug catcher oil outlet flow	GOOD
4		0201LIC0231_02.PV	Slug catcher drum level High SP	SUSPECT
5		0201LIC0231_02B.PV	Slug catcher drum level Low SP	GOOD
6		0201FI0231_01.PV	Slug catcher drum gas flow	GOOD
7	Train-3 (1st Stage Separator)	0201LIC0331_03.PV	Separator Oil level	GOOD

S.No	Section	Instrument Tag name	Description	Instrument Status
8		0201FIC0331_03.PV	Separator Oil outlet flow	GOOD
9		0201FIC0331_06.PV	Separator gas outlet flow	GOOD
10		0201FI0331_02.PV	Separator water outlet flow	GOOD
11		0201LIC0331_01.PV	Separator water level	GOOD
12	Train-3 (2nd Stage Separator)	0201LIC0431_04.PV	Separator Oil level	GOOD
13		0201LIC0431_02.PV	Separator Water level	GOOD
14		0201TIC0431_06.PV	Separator temperature control	GOOD
15		0201FI0431_02.PV	Separator gas outlet flow	GOOD
16		0201FI0431_04.PV	Separator water outlet flow	GOOD
17	Train-3 (1st Stage Desalter)	0201LIC1231_64.PV	Desalter Interphase level	GOOD
18		0201FIC1231_64.PV	Fresh water feed flow	GOOD
19		0201FI1231_63.PV	Desalter water flow to Degasser	GOOD
20		0201EI1231_61.PV	Transformer Volt	GOOD
21		0201II1231_61.PV	Transformer Current	GOOD
22	Train-3 (2nd Stage Desalter)	0201LIC1232_64.PV	Desalter Interphase level	GOOD
23		0201EI1232_61.PV	Transformer Volt	GOOD
24		0201II1232_61.PV	Transformer Current	GOOD
25	Train-3 (Oil Stabilizer column)	0201TIC1431_01.PV	Stab column Reboiler-1 Temp	GOOD
26		0201FIC1431_01.PV	Stab column Reboiler-1 HO flow	GOOD
27		0201TIC1432_01.PV	Stab column Reboiler-2 Temp	GOOD
28		0201FIC1432_01.PV	Stab column Reboiler-2 HO flow	GOOD
29		0201LIC1431_06.PV	Stab column btm level	GOOD
30		0201FI1431_02.PV	Stab column oil feed flow	GOOD
31		0201FI1431_06.PV	Stab column gas outlet flow	GOOD
32		0201DPI1431_02.PV	Stab column Diff. pressure	GOOD
33		0201TI1431_11.PV	Stab column ovhd temperature	GOOD
34		0201TI1431_02.PV	Stab column temperature	GOOD
35		0201TI1431_04.PV	Stab column temperature	GOOD
36		0201TI1431_05.PV	Stab column temperature	GOOD
37		0201TI1431_07.PV	Stab column temperature	GOOD
38		0201TI1431_08.PV	Stab column temperature	GOOD
39		0201TI1431_09.PV	Stab column temperature	GOOD
40		0201TI1431_16.PV	Stab column btm draw temperature	GOOD
41		0201FI0831_01.PV	Oil production flow to FST	GOOD
42	Train-4 (Slug catcher)	0201FIC0241_02.PV	Slug catcher oil outlet flow	SUSPECT

S.No	Section	Instrument Tag name	Description	Instrument Status
43		0201LIC0241_02.PV	Slug catcher drum level High SP	GOOD
44		0201LIC0241_02B.PV	Slug catcher drum level Low SP	GOOD
45		0201FI0241_01.PV	Slug catcher drum gas flow	GOOD
46	Train-4 (1st Stage Separator)	0201LIC0341_03	Separator Oil level	GOOD
47		0201FIC0341_03	Separator Oil outlet flow	GOOD
48		0201FIC0341_06	Separator gas outlet flow	GOOD
49		0201FI0341_02.PV	Separator water outlet flow	GOOD
50		0201LIC0341_01.PV	Separator water level	GOOD
51	Train-4 (2nd Stage Separator)	0201LIC0441_04.PV	Separator Oil level	GOOD
52		0201LIC0441_02.PV	Separator Water level	GOOD
53		0201TIC0441_06.PV	Separator temperature control	GOOD
54		0201FI0441_02.PV	Separator gas outlet flow	GOOD
55		0201FI0441_04.PV	Separator water outlet flow	GOOD
56	Train-4 (1st Stage Desalter)	0201LIC1241_64.PV	Desalter Interphase level	GOOD
57		0201FIC1241_64.PV	Fresh water feed flow	GOOD
58		0201FI1241_63.PV	Desalter water flow to Degasser	GOOD
59		0201EI1241_61.PV	Transformer Volt	GOOD
60		0201II1241_61.PV	Transformer Current	GOOD
61	Train-4 (2nd Stage Desalter)	0201LIC1242_64	Desalter Interphase level	GOOD
62		0201EI1242_61.PV	Transformer Volt	GOOD
63		0201II1242_61.PV	Transformer Current	GOOD
64	Train-4 (Oil Stabilizer column)	0201TIC1441_01.PV	Stab column Reboiler-1 Temp	GOOD
65		0201FIC1441_01.PV	Stab column Reboiler-1 HO flow	GOOD
66		0201TIC1442_01.PV	Stab column Reboiler-2 Temp	GOOD
67		0201FIC1442_01.PV	Stab column Reboiler-2 HO flow	GOOD
68		0201LIC1441_06.PV	Stab column btm level	GOOD
69		0201FI1441_02.PV	Stab column oil feed flow	GOOD
70		0201FI1441_06.PV	Stab column gas outlet flow	GOOD
71		0201DPI1441_02.PV	Stab column Diff. pressure	GOOD
72		0201TI1441_11.PV	Stab column ovhd temperature	GOOD
73		0201TI1441_02.PV	Stab column temperature	GOOD
74		0201TI1441_04.PV	Stab column temperature	GOOD
75		0201TI1441_05.PV	Stab column temperature	GOOD
76		0201TI1441_07.PV	Stab column temperature	GOOD
77		0201TI1441_08.PV	Stab column temperature	GOOD

S.No	Section	Instrument Tag name	Description	Instrument Status
78		0201TI1441_09.PV	Stab column temperature	GOOD
79		0201TI1441_16.PV	Stab column btm draw temperature	GOOD
80		0201FI0841_01.PV	Oil production flow to FST	GOOD
81	Hot Oil Heater	0201TIC6091_04.PV	Heater COT	GOOD
82		0201FIC6091_52.PV	FG flow to burner	GOOD
83		0201TI6091_01.PV	Hot oil return temperature	GOOD
84		0201AIC6091_52.PV	Heater Excess O2 Analyzer	SUSPECT
85		0201AI6091_51.PV	Heater CO analyzer	SUSPECT
86		0201FIC2892_52.PV	Glycol reboiler Hot oil flow	GOOD

Mode change testing for the potential APC MVs. This testing involved temporarily changing the operating mode of a PID loop to assess different modes in relation to APC applications. The loops were put into test mode, and either the setpoint (AUTO) or output (MAN) values were adjusted within the range specified in operating envelop.

Step Test for potential APC MVs. The primary objective of conducting the step test is to establish the permissible movement range for each identified manipulated variable (MV) that yields an adequate response in the relevant controlled variables (CVs). If none of the targeted CVs exhibit a response, the step size will be increased accordingly. During the pretest visit, small step changes were made to find optimal responses from each CV. This process was repeated for every MV to determine appropriate step sizes.

Following comprehensive data analysis, simulations, site inspections, and validation of engineering documentation, PID loops were identified as potential manipulated variables (MV) for the APC Model, as detailed in [Table-5](#), with conservatively determined allowable step sizes.

Table 5—APC Model Potential Manipulated Variables (MV)

S. No	Section	Loop name	Description	Allowed Step size	
				Min	Max
1	Train-3 (Slug catcher)	0201FIC0231_02.OP	Slug catcher oil outlet flow	±1%	±2%
2	Train-3 (1st Stage Separator)	0201FIC0331_03.SP	Separator Oil outlet flow	±5000 BPD	±10000 BPD
3	Train-3 (2nd Stage Separator)	0201LIC0431_04.OP	Separator Oil level	±2%	±4%
4	Train-3 (1st Stage Desalter)	0201FIC1231_64.SP	Fresh water feed flow	±500 kg/hr	±800 kg/hr
5	Train-3 (Oil Stabilizer column)	0201TIC1432_01.SP	Stab column Reboiler-2 Temp	±2°C	±10°C
6		0201FIC1432_01.SP	Stab column Reboiler-2 HO flow	±5000 BPD	±10000 BPD
7	Train-4 (Slug catcher)	0201FIC0241_02.OP	Slug catcher oil outlet flow	±1%	±2%
8	Train-4 (1st Stage Separator)	0201FIC0341_03.SP	Separator Oil outlet flow	±5000 BPD	±10000 BPD
9	Train-4 (2nd Stage Separator)	0201LIC0441_04.OP	Separator Oil level	±2%	±4%
10	Train-4 (1st Stage Desalter)	0201FIC1241_64.SP	Fresh water feed flow	±500 kg/hr	±800 kg/hr
11	Train-4 (Oil Stabilizer column)	0201TIC1442_01.SP	Stab column Reboiler-2 Temp	±2°C	±10°C

S. No	Section	Loop name	Description	Allowed Step size	
				Min	Max
12		0201FIC1442_01.SP	Stab column Reboiler-2 HO flow	±5000 BPD	±10000 BPD
13	Hot Oil Heater	0201TIC6091_04.SP	Heater COT	±2°C	±5°C

APC functional design specifications

The APC Application achieves the objectives by appropriate adjustments of selected process independent variables that are referred to as Manipulated Variables (MVs). This manipulation is performed by the APC controller. The controller calculates optimum process settings based on the process model, production targets, and readings of current values of selected process dependent variables that are referred as Controlled Variables (CVs). APC also reads variables referred to as Disturbance Variables (DVs) or Feed-Forward variables (FFWs), by accounting for changes of such measured disturbances, the APC controller is able to take proactive action that minimizes the impact of such changes on the process.

After data analysis, identification of MVs and pre-testing; the APC solution functional design specifications were developed in which all areas including Fallback strategy, interface philosophy, Logic charts and cybersecurity requirements were detailed. APC model identification is based on 2 algorithms, one being the traditional Finite Impulse Response (FIR) algorithm and the other, the Subspace algorithm. The APC controller always maintains MVs within their limits and these limits shall be defined as Engineering limits and Operating limits (within Engineering limits, set by Operators) and adjusted in line with the process safety and operational requirements. Following are the APC controllers deployed as part of the project scope.

- Main APC Controller – **CON_APC**
- Train-3 APC Sub-Controller – **T3_APC**
- Train-4 APC Sub-Controller – **T4_APC**
- Hot Oil Heater APC Sub-Controller – **HO_APC**

APC Model Variables Mapping

Each APC controller is tailored to serve specific sections of the plant, with both main and sub-controllers working in tandem to optimize performance. The Main APC Controller (CON_APC) oversees the integrated process, while the train-3 (T3_APC) and Train-4 (T4_APC) sub-controllers manage their respective process trains. For each controller, a detailed mapping of Manipulated, Controlled, and Disturbance Variables was established as listed in [Table-6](#).

Table 6—T3/T4 APC Model Variables Mapping

S.No	Variable Name	Units	Description	Type	Optimization Strategy	Control narrative
MANIPULATED VARIABLES (MVs)						
1	0201FIC0231_02.OP	%	Fluid outlet OP from Slug catcher	To control Ramp CV (LIC0231 02)	Maintain	Maintain this flow to control the level of Slug catcher. This flow is very critical to stabilize the level when slug disturbs the process.
2	0201FIC0331_03.SP	BPD	Oil outlet flow from 1st Stage Separator	To control Ramp CV (LIC0331 03)	Maintain	Maintain this level to stabilize flow to 2nd stage separator

S.No	Variable Name	Units	Description	Type	Optimization Strategy	Control narrative
						which help stabilizes Oil stabilizer column profile.
3	0201LIC0431_04.OP	%	2nd Stage Separator level Output OP	To control Ramp CV (LIC0431 04)	Maintain	Maintain this level to stabilize the Oil Stabilizer column profile.
4	0201FIC1231_64.SP	kg/hr	Fresh water flow to Desalter	To control Stable Ratio CV	MINimize	MINimize this flow to optimize water to oil ratio subject to Salt content spec. in Crude oil
5	0201FIC1431 01.SP	BPD	Stab column Reboiler-1 HO flow	To control	MINimize	MINimize the Stabilizer column
6	0201FIC1432 01.SP	BPD	Stab column Reboiler-2 HO flow	StableCVs		produce On Spec Crude oil product at the lowest energy consumption.
DISTURBANCE VARIABLES (MVs)						
1	0201PI1391_04.PV	Barg	Production Manifold pressure TL-3	This DV has potential to detect slugs upstream of Slug catcher.		
2	0201PI1392_04.PV	Barg	Production Manifold pressure TL-4			
3	0201TIC6091 04.PV	Deg.C	Heater Outlet COT	Heater COT directly affects the Stabilizer column heat duty.		
CONTROLLED VARIABLES (CVs)						
1	0201LIC0231 02.PV	%	Slug catcher fluid level	Feed forward type	Maintain	Maintain this level to reduce disturbances in downstream drums
2	0201FIC0231 02.PV	BPD	Slug catcher outlet flow	Feed forward type	Maintain	Maintain this flow within range to make adjustments for slug disturbances.
3	0201LIC0331 03.PV	%	1st Stage Separator level	Feed forward type	Maintain	Maintain this level to reduce disturbances in 2nd stage separator
4	0201LIC0431_04.PV	%	2nd Stage Separator level	Feed forward type	Maintain	Maintain this level to reduce disturbances in Stabilizer column
	T3_WORATIO		Wash Water to Oil ratio	Feed back type	MINimize	Minimize this ratio to reduce wash water consumption based on Oil to Desalter feed rate
6	Salt in crude (Inferential)	ptb	Salt content in Crude oil	Feed back type	Maintain <10	Maintain this quality as per required spec at minimum wash water required.
7	0201TIC1431_01.PV	Deg.C	Oil stab. Column reboiler-1 temperature	Feed forward type	MINimize	Maintain this temperature at lowest possible margin to produce On spec Crude oil. Only 1 of the reboilers will be in line at a time.
8	0201TIC1432_01.PV	Deg.C	Oil stab. Column reboiler-2 temperature	Feed forward type	MINimize	

S.No	Variable Name	Units	Description	Type	Optimization Strategy	Control narrative
9	TVP (Inferential)	psi	TVP content of Crude oil at Stab oil cooler outlet	Feed forward type	Maintain 10 ± 0.2	Maintain this quality as per required spec at minimum energy consumption.
10	H2S (Inferential)	ppm	H2S content of Crude oil	Feed forward type	Maintain <35	Maintain this quality as per required spec at minimum energy consumption.

Feedback type – APC controls the CV based on feedback response from plant inputs. This means the corresponding MV will start moving only after the CV is predicted to be touching/crossing limits.

Feed forward type – APC controls the CV based on feedback along with feed forward disturbance input from plant. This means the corresponding MV will start moving based on the CV prediction from feedback response as well as CV prediction from disturbance model associated with it.

Furthermore, the APC variables were categorized into Independent Variables and Dependent Variables and were configured in APC controller as shown in [Picture-3](#) and [Picture-4](#).

Inputs	Description	Units	Service Request	Measurement	Operator Low Limit	Operator High Limit	Critical
► FIC0231_02.O	T3 Slug Catcher outlet Flow OP	%	On	69.83	58	70	No
FIC0331_03.S	T3 1st Stg Separator outlet flow	BPD	On	67249.3	35000	85000	No
LIC0431_04.O	T3 2nd Stg Separator outlet valve OP	%	On	41.79	30	48	No
FIC1231_64.S	T3 Wash water feed flow	kg/hr	On	4500	3200	4500	No
FIC1431_01.S	T3 Stab column Reboiler-1 HO flow	BPD	Off	26012.6	30000	1.2E+05	No
FIC1432_01.S	T3 Stab column Reboiler-2 HO flow	BPD	On	63275.4	30000	1.2E+05	No
PI1391_04.PV	Trunkline-3 Pressure DV	barg	On	16.92			No
PI1392_04.PV	Trunkline-4 Pressure DV	barg	On	16.01			No
TIC6091_04.P	Hot oil Heater COT DV	C	On	202.4			No

Picture-3—Independent Variables of T3 APC Controller

Outputs	Description	Units	Service Request	Measurement	Operator Low Limit	Operator High Limit	Ramp Setpoint	Model Prediction	Engineer Request
► LIC0231_02.P	T3 Slug catcher level	%	On	46.85	35	50	41	11.66	On
FIC0231_02.P	T3 Slug catcher outlet flow	BPD	On	73214.1	60000	95000		31937.6	On
LIC0331_03.P	T3 1st Stg Separator level	%	On	42.65	42	43	42	43.4	On
LIC0431_04.P	T3 2nd Stg Separator level	%	On	42.77	42	43	42	43.41	On
T3_WORATIO	T3 Water to Oil ratio	ratio	On	1.38	1.4	2		1.19	On
T3_SALT	Salt content in Crude oil (Inf)	ptb	On	5.038	0	12		4.059	On
TIC1431_01.P	T3 Stab column Reboiler-1 Temp	C	Off	41.18	148	152		0	On
TIC1432_01.P	T3 Stab column Reboiler-2 Temp	C	On	150.3	148	152		149	On
T3_TVP	T-3 Crude oil TVP (Inf)	psi	On	9.82	9	10		9.852	On
T3_H2S	H2S content in Crude oil (Inf)	ppm	On	2.998	0	15		3.015	On

Picture-4—Dependent Variables of T3 APC Controller

APC Variables for controller HO_APC

Similarly, variable mapping was done for Hot Oil APC controller as shown in [Table-6a](#), [Picture-5](#) and [Picture-6](#).

Table 6a—APC Hot Oil (HO_APC) Controller Variables Mapping

No.	Variable Name	Units	Description	Type	Optimiz ation Strategy	Control narrative
MANIPULATED VARIABLES						
1	0201TIC6091_04.SP	Deg.C	Heater Outlet COT	To control Stable CV	MINimize	MINimize this temperature to supply heat to the process side reboilers at lowest possible margin. This also helps to minimize Fuel gas consumption.
DISTURBANCE VARIABLES						
1	0201PI6091 64.PV	Barg	FG pressure	This DV could affect the Heater outlet COT.		
2	MAX_REB_TEMP	Deg.C	Max of all 4 Reboiler temperatures	This calculation DV informs about the disturbance caused by any fluctuation in any reboiler temperature to Hot oil system.		
CONTROLLED VARIABLES						
1	TC6091_04.PV	Deg.C	Heater COT PV	Feed forward type	Maintain	Maintain the COT PV within limits
2	FIC6000_21.PV	BPD	Hot oil circulation flow	Feed back type	Maintain	Maintain this flow within limits to ensure minimum flow available for circulation.
3	MAX SKIN TEMP (Calculated CV)	Deg.C	Max skin temperature of Heater coils	Feed forward type	Maintain	Maintain the COT w.r.t honouring this constraint CV
4	0201FIC1431_01.O P	%	T-3 Stab reboiler-1 HO flow OP	Feed back type	Constraint	When the HO flow OP reaches a saturation point at process sides, Heater COT shall increase to provide heat duty margin to in-line reboilers
5	0201FIC1432_01.O P	%	T-3 Stab reboiler-2 HO flow OP	Feed back type	Constraint	
6	0201FIC1441_01.O P	%	T-4 Stab reboiler-1 HO flow OP	Feed back type	Constraint	
7	0201FIC1442_01.O P	%	T-4 Stab reboiler-2 HO flow OP	Feed back type	Constraint	

Inputs	Description	Units	Service Request	Measurement	Operator Low Limit	Operator High Limit	Critical
► TIC6091_04.S	Hot oil Heater COT Setpoint	C	On	 203	203	205	No
PI6091_64.PV	Hot oil Heater Fuel gas Pressure DV	barg	On	4.698			No
MAX_REB_TEMP	Max Reboiler temperature DV	C	On	150.3			No

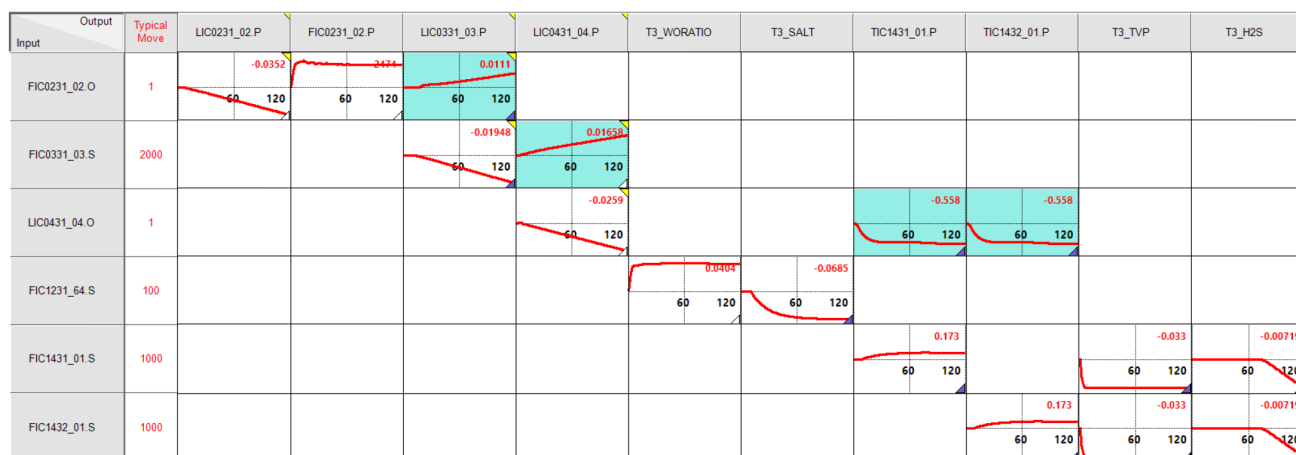
Picture-5—Independent Variables of HO_APC Controller

Outputs	Description	Units	Service Request	Measurement	Operator Low Limit	Operator High Limit	Ramp Setpoint	Model Prediction	Engineer Request
TC6091_04.PV	Hot oil heater COT PV	C	On	202.4	195	205		203.1	On
FIC6000_21.P	Hot Oil recirculation Flow		On	3.64E+05	3E+05	4.5E+05		3.652E+05	On
MAX_SKIN_T	Hot Oil Heater Max Skin temperature	C	On	287.4	200	300		290.7	On
FIC1431_01.O	T3 Stab column Reb-1 valve OP	%	Off	0	0	98		0	On
FIC1432_01.O	T3 Stab column Reb-2 valve OP	%	On	24.97	5	98		26.27	On
FIC1441_01.O	T4 Stab column Reb-1 valve OP	%	On	19.68	5	98		21.99	On
FIC1442_01.O	T4 Stab column Reb-2 valve OP	%	Off	0	0	98		0	On

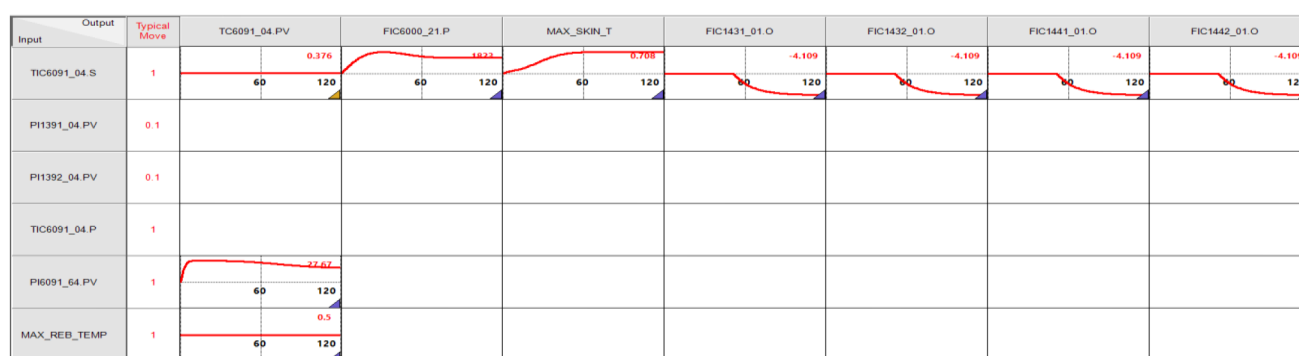
Picture-6—Dependent Variables of HO_APC Controller

Gain Matrix for the Implemented APC controllers

The process model transfer function matrix for oil train APC and Hot Oil APC was developed in APC software after multiple rounds of simulations as is shown in [Trend-1](#) and [Trend-2](#).



Trend-1—Process Model Transfer Function Matrix with sub-model gains for T3_APC



Trend-2—Process Model Transfer Function Matrix with sub-model gains for HO_APC

INFERENCE QUALITIES (IQ) / Soft Sensors

An Inferential Quality sensor is a mathematical relation that calculates or predicts a property using other available process data. IQ will reduce process-deadtime, provide redundancy for lab analysis results and provide predictions in real time. These inferred quality variables are then used as Controlled variables in APC and set a control range for the operators to adjust in the APC HMI. The IQs developed for this project are listed in [Table-7](#) with respective inferential equations.

Table 7—Project Inference / Soft-sensors

S. No	Section	Inferential name	Description	Model Equation
1	Train-3	T3_TVP	Crude oil TVP (psi) at Stabilizer oil cooler outlet	$T3_RVP_static = + 24.9228 - 0.0616321 * PIC1431_09.OP - 0.081277 * TI1431_07.PV - 0.0777809 * TI1431_11.PV$
2		T3_H2S	Crude oil H2S (ppm) at Stabilizer oil cooler outlet	$T3_H2S_static = + 32.3609 - 0.640196 * PIC0431_13.PV - 0.00121855 * FIC1232_61.PV - 0.0152265 * TI1431_02.PV$
3		T3_Salt	Crude oil Salt content (ptb) at Stabilizer oil cooler outlet	$T3_Salt_static = -24.8633 + 0.480937 * TI0432_13.PV - 0.966828 * III231_61.AI.PV + 0.000931079 * FIC1231_64.PV - 4.14877 * DPIC1232_62.PV$
4	Train-4	T4_TVP	Crude oil TVP (psi) at Stabilizer oil cooler outlet	$T4_RVP_static = + 24.9228 - 0.0616321 * PIC1441_09.OP - 0.081277 * TI1441_07.PV - 0.0777809 * TI1441_11.PV$
5		T4_H2S	Crude oil H2S (ppm) at Stabilizer oil cooler outlet	$T4_H2S_static = -14.33 - 0.066447 * PIC0441_13.OP + 0.248945 * TI1241_64.PV + 0.00576539 * FIC1442_01.OP$
6		T4_Salt	Crude oil Salt content (ptb) at Stabilizer oil cooler outlet	$T4_Salt_static = + 8.77747 - 0.282771 * TIC0441_06.PV - 2.35325 * III242_61.AI.PV + 84.561 * DPI1441_02.PV + 0.121442 * TI1441_11.PV$

To enable continuous monitoring of these inferential points, numeric points in DCS were created and APC application writes the inferential prediction result to these DCS points.

Network Architecture & DCS – APC Interface

The architecture and interface logics were developed to establish interface between DCS and APC while complying with all applicable cyber security requirements as briefed in this section. The implemented architecture is shown in Figure-2 and explained as:

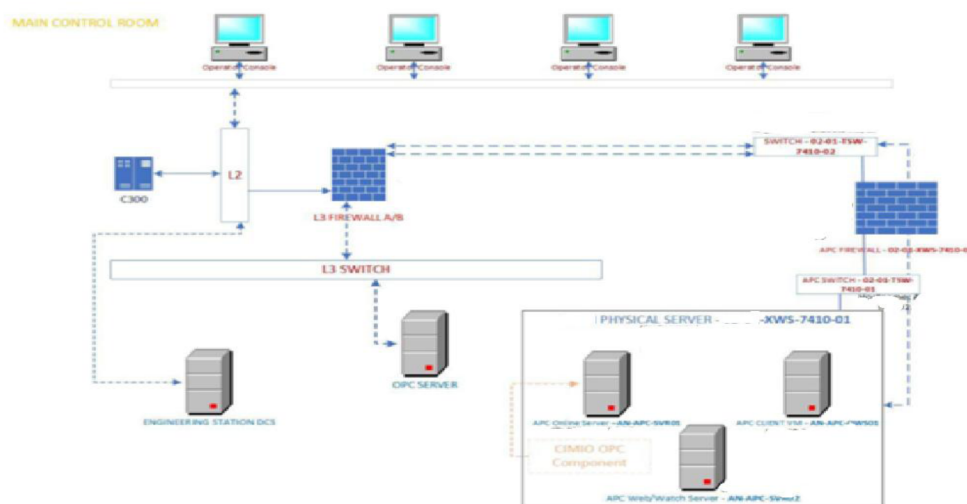


Figure 2—APC-DCS Interface Architecture

- A physical server which hosts the 3 required APC components as explained below.
- **APC Online Server VM** – Includes APC and Inferential Qualities Controllers. The Online components of APC Controller consist of the Control, APC Manage, and OPC Server.
- **APC Web/Watch Server VM** – Production Control Web Server is an online, web-based user interface.
- **APC client VM**- Serves as the Dedicated APC operating console for the operators through web browsers to access PCWS screens.
- The Physical server was connected to L3 network switch through APC switch and Honeywell switch through APC firewall.
- The OPC component was installed in Aspen Online server (AN-APC-SVR01) and communication was established with DCS OPC server through OPC-DA protocol. Necessary DCOM configurations were configured.
- The APC Firewall AN-APC-FGFW is licensed with Industrial grade and configured with enabling OPC-DA service only, with no restrictions on ports. This will ensure that only OPC-DA data requests shall pass through the firewall by taking necessary dynamic ports while the other communication protocols will be blocked. The same configuration was ensured in the existing L3 Firewall as well.
- The Web/Watch server VM will host the Operator displays for controlling and operating APC variables. This shall connect to Online server and publishes the dashboards through Web browser in Operator stations and the Workstation AN-APC-OWS01.

An APC Overview graphic page containing APC On/Off status and MV status change graphic works was configured in DCS for operator convenience as shown in [Figure-3](#).

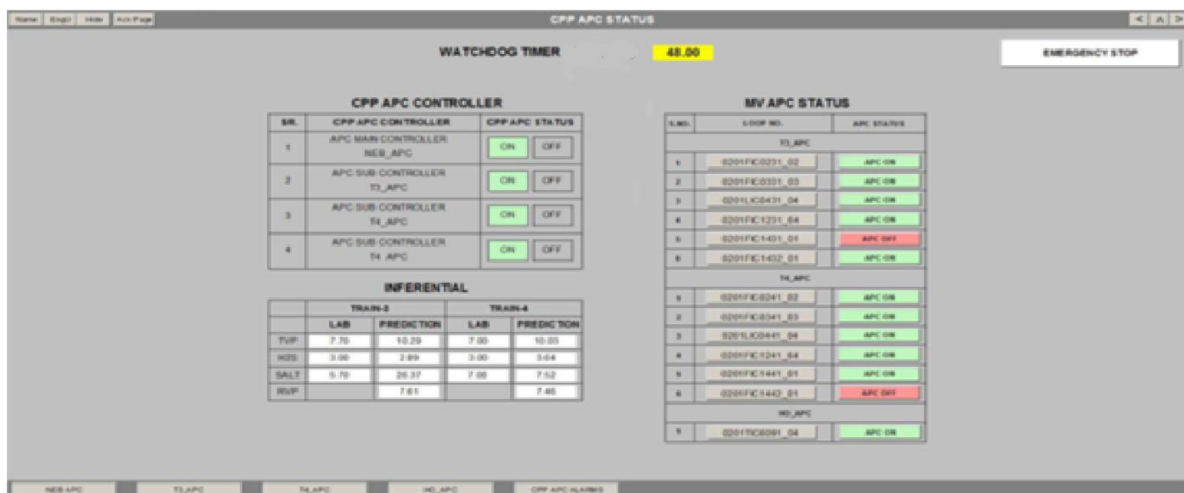


Figure 3—APC – DCS Interface Graphics

DCS Configuration

Following tasks were carried out in the DCS to implement the APC solution in closed loop:

- DCS points were build for APC on the Train-3/4 respective DCS Controller: Intermediate points were configured per Main controller, sub controllers and MVs.
- Modify MV points which are present in DCS controller (Download of controllers carried out while they were online for each loop with extra precaution, with support from Operations & Maintenance teams).

Manipulated Variables categorization. Based on the category of the Manipulated Variable the corresponding PID loop in the DCS were modified. The categories are as follows:

- **Category 1:** The APC writes SP to the PID loop and the PID loop is stand alone in the DCS. i.e. it does not have any master PID point
- **Category 2:** The APC writes SP to the PID loop and the PID loop has a master PID loop in the DCS from which it can receive the set point
- **Category 3:** The APC will write OP to the PID loop

The MVs modified for this project are listed in [Table-8](#).

Table 8—Manipulated Variables Categories

S. No	Section	Loop name	Description	MV type
1	Train-3 (Slug catcher)	0201FIC0231_02.OP	Slug catcher oil outlet flow OP	Category 3
2	Train-3 (1st Stage Separator)	0201FIC0331_03.SP	Separator Oil outlet flow	Category 2
3	Train-3 (2nd Stage Separator)	0201LIC0431_04.OP	Separator Oil level OP	Category 3
4	Train-3 (1st Stage Desalter)	0201FIC1231_64.SP	Fresh water feed flow	Category 1
5	Train-3 (Oil Stabilizer column)	0201FIC1431_01.SP	Stab column Reboiler-1 HO flow	Category 2
6	Train-3 (Oil Stabilizer column)	0201FIC1432_01.SP	Stab column Reboiler-2 HO flow	Category 2
7	Train-4 (Slug catcher)	0201FIC0241_02.OP	Slug catcher oil outlet flow OP	Category 3
8	Train-4 (1st Stage Separator)	0201FIC0341_03.SP	Separator Oil outlet flow	Category 2
9	Train-4 (2nd Stage Separator)	0201LIC0441_04.OP	Separator Oil level OP	Category 3
10	Train-4 (1st Stage Desalter)	0201FIC1241_64.SP	Fresh water feed flow	Category 1
11	Train-4 (Oil Stabilizer column)	0201FIC1441_01.SP	Stab column Reboiler-1 HO flow	Category 2
12	Train-4 (Oil Stabilizer column)	0201FIC1442_01.SP	Stab column Reboiler-2 HO flow	Category 2
13	Hot Oil Heater	0201TIC6091_04.SP	Heater COT	Category 1

Interface points

The Hierarchy of Intermediate Points is shown in [Figure-4](#).

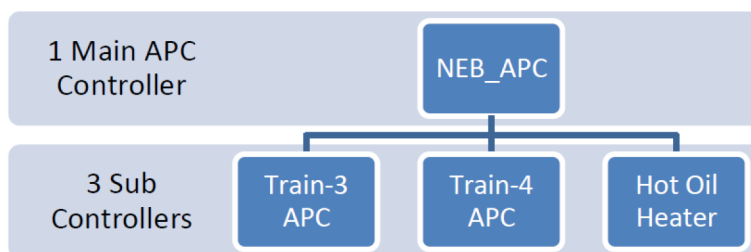


Figure 4—APC Controllers Hierarchy

Each controller has the following interface points:

- One MAIN controller point per controller
- One SUB controller point per sub-controller
- One MV interface point per manipulated variable

All 3 sub controllers were configured to execute at 1 minute frequency.

WATCHDOG TIMER and ON/OFF FLAG BLOCK

The Watchdog timer and ON/OFF Status blocks are components within the Main Template block. The Timer Resetcmd and Strtstop values are transmitted from the APC online to the DCS, which initiates a watchdog timer. While the APC is operational and communication with the DCS is maintained, the timer continues to run. If communication is interrupted, the timer will expire, causing the APC controller to shut down and the MV tags to be shed.

When the ONREQ value is set to ON (set to 1.0), the MAIN block will RESET and START the CTLOM timer. If the APC controller is not turned ON in the specified time (CTLOM_CNTDOWN = 240 seconds) or if the communication watchdog (CNTDOWN = 1.5 times the controller cycle) times out, then the MAIN block's ONREQ will be turned OFF (set to 0.0).

The ONSTS when set will stop the timer, indicating that APC ON status is ON. If the ON Status is NOT set in a specified amount of time the CTLOM timer will time out, forcing the ONREQ to turn OFF and shed all MVs, and turning OFF all subcontrollers that may be ON.

DCS Fallback Logics & Alarms

APC variables fallback logics were configured in DCS for a safe and smooth handover of control to DCS/operator during predefined events including communication failure, operating envelope thresholds and manual initiation by operator from the console. The fallback strategy is briefed in [Table-9](#).

Table 9—APC-DCS Interface Fallback Strategy

S.No	Loop name	Description	DCS Mode / Fallback Mode	APC Mode
1	0201FIC0231_02	Slug catcher oil outlet flow	OPERATOR AUTO	REMOTE MANUAL
2	0201FIC0331_03	Separator Oil outlet flow	OPERATOR CASCADE	REMOTE AUTO
3	0201LIC0431_04	Separator Oil level OP	OPERATOR AUTO	REMOTE MANUAL
4	0201FIC1231_64	Fresh water feed flow	OPERATOR AUTO	REMOTE AUTO
5	0201FIC1431_01	Stab column Reboiler-1 HO flow	OPERATOR CASCADE	REMOTE AUTO
6	0201FIC1432_01	Stab column Reboiler-2 HO flow	OPERATOR CASCADE	REMOTE AUTO
7	0201FIC0241_02	Slug catcher oil outlet flow	OPERATOR AUTO	REMOTE MANUAL
8	0201FIC0341_03	Separator Oil outlet flow	OPERATOR CASCADE	REMOTE AUTO
9	0201LIC0441_04	Separator Oil level	OPERATOR AUTO	REMOTE MANUAL
10	0201FIC1241_64	Fresh water feed flow	OPERATOR AUTO	REMOTE AUTO
11	0201FIC1441_01	Stab column Reboiler-1 HO flow	OPERATOR CASCADE	REMOTE AUTO
12	0201FIC1442_01	Stab column Reboiler-2 HO flow	OPERATOR CASCADE	REMOTE AUTO
13	0201TIC6091_04	Heater COT	OPERATOR AUTO	REMOTE AUTO

When APC is turned ON by the operator, the PID loop goes into corresponding APC Mode as shown in [Table-9](#). The DCS mode is the usual operating mode and this will also serve as the Fallback mode during the conditions below:

- When APC turns OFF
- When APC-DCS communication failure happens
- When any of the configured MV returns bad quality value from DCS field instrument.

Following alarm configurations were made to ensure maximum uptime of APC controller:

- **CONTROLLER DROP ALARM:** DCS Alarm configured for any APC Sub controller or CON_APC Main controller turns OFF due to operator action or communication failure.
- **MV DROP ALARMS:** DCS Alarm configured for any MV is dropped from APC by operator action or due to communication failure or bad PV quality. Then it goes into its configured fallback mode.
- **CV DEVIATION ALARMS:** DCS Alarm was created for CV deviation. The deviation of CV from its high or low limit will be monitored and whenever it crosses, a timer of 15 minutes will be activated. If the CV does not come back in range within 15 min and remain outside the limit, then this alarm will be triggered in DCS as shown in [Figure-5](#).

CPP APC LOW & HIGH LIMIT DEVIATION ALARM

SR.	LOOP NO.	HI DEV ALM	LO DEV ALM		SR.	LOOP NO.	HI DEV ALM	LO DEV ALM	
1	0201LIC0231_02.PV_APC				14	0201TC6091_04.PV_APC			
2	0201FIC0231_02.PV_APC				15	0201FIC6000_21.PV_APC			
3	0201LIC0331_03.PV_APC				16	0201FIC1431_01.OP_APC			
4	020LIC0431_04.PV_APC				17	0201FIC1432_01.OP_APC			
5	T3_WORATIO_APC				18	0201FIC1441_01.OP_APC			
6	0201TIC1431_01.PV_APC				19	0201FIC1442_01.OP_APC			
7	0201TIC1432_01.PV_APC				20	T3_TVP_APC			
8	0201LIC0241_02.PV_APC				21	T3_SALT_APC			
9	0201FIC0241_02.PV_APC				22	T3_H2S_APC			
10	0201LIC0341_03.PV_APC				23	T4_TVP_APC			
11	0201LIC0441_04.PV_APC				24	T4_SALT_APC			
12	0201TIC1441_01.PV_APC				25	T4_H2S_APC			
13	0201TIC1442_01.PV_APC				26	MAX_SKIN_TEMP_APC			
					27	T4_WORATIO			

Figure 5—APC CV Deviation Alarm Graphics Developed in DCS for this project

Results & Observations / Post commissioning Evaluation

The implementation of Advanced Process Control (APC) at Central Processing facilities yielded significant benefits, demonstrating both operational excellence and financial gains. APC has seamlessly integrated into the existing processes, optimizing production and enhancing efficiency.

Key Highlights

- **Increased Production:** Through coordinated control, APC has achieved a notable increase in oil recovery rates. APC has led to more stable and consistent process operations, reducing variability and enhancing control over key parameters.
- **Energy Savings:** APC has contributed to significant energy savings by optimizing energy-intensive processes such as Hot oil Heater.
- **Environmental Impact:** With APC, a reduction in emissions was achieved to minimize our environmental footprint.

Evaluation period

One Month (31 days) – 11th December to 11th January.

Evaluation Methodology

Compare the KPI data when controller is running with the baseline data as mentioned in section 2.2.

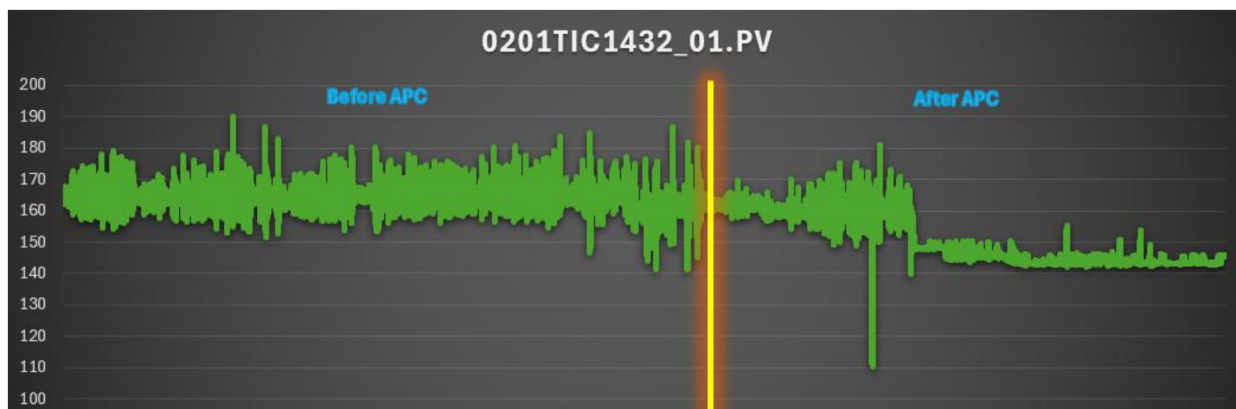
A summary of the realized benefits resulting from the deployment of APC in upstream GOSP trains, as outlined within the scope of this initiative and project, is provided in [Table 10](#).

Table 10—APC Deployment in upstream GOSP Trains – Realized Benefits

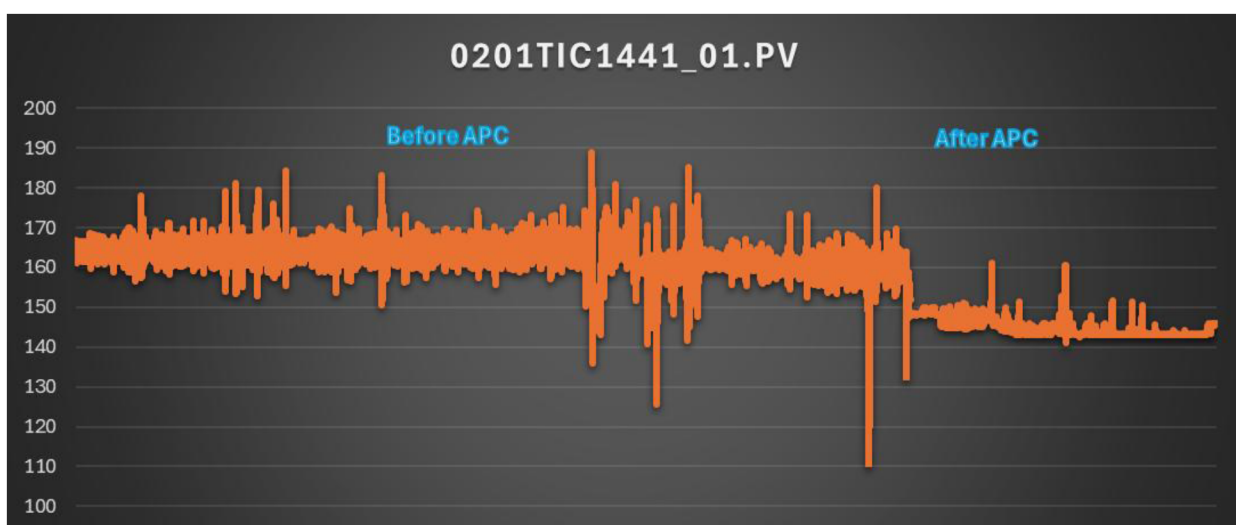
S. No	Description	Train	Baseline	Evaluation	UOM	KPI Target	KPI Achieved	% Improvement
1	Maximize oil Production	T3	94.2	95.58	%	1-2%	1.381	1.336%
		T4	94.4	95.69		1-2%	1.291	
2	Maintain TVP (True Vapor Pressure) of the stabilized crude within 10 psi +/- 0.2 psi.	T3	7.77	10.06	psi	9.8-10.2	10.05	29.34 % SD
		T4		10.23		9.8-10.2		
3	Optimize Lean Gas Consumption	Common	24.49	23.25	sm3/hr/MBOPD	7-15%	5.06	5.06%
4	Slug Disturbance Rejection	T3 Level	4.44	5.75	%	5-10%		29.64 (% SD)
		T3 Flow	9782	11960	BPD			22.36 (% SD)
		T4 Level	4.035	3.4	%			13.94 (% SD)
		T4 Flow	14220	6376	BPD			55.05 (% SD)
5	Column Stabilization	T3	4.38	1.898	Deg C	5-10%		56.51 (% SD)
		T4	3.8	2.309				39.01 (% SD)
6	Wash Water	T3	1.291	0.8163	%	1 – 3 %	36.77%	35.46%
		T4	1.241	0.8172			34.15%	
7	Maintain H2S	T3	0.478	0.089	ppm	below 35 ppm	0.085	81.37% (SD)
		T4		0.08				
8	Maintain Salt Content	T3	1.598	1.19	PTB	< 10	1.227	23.22% (SD)
		T4		1.26				

APC Performance & Model Performance Analysis

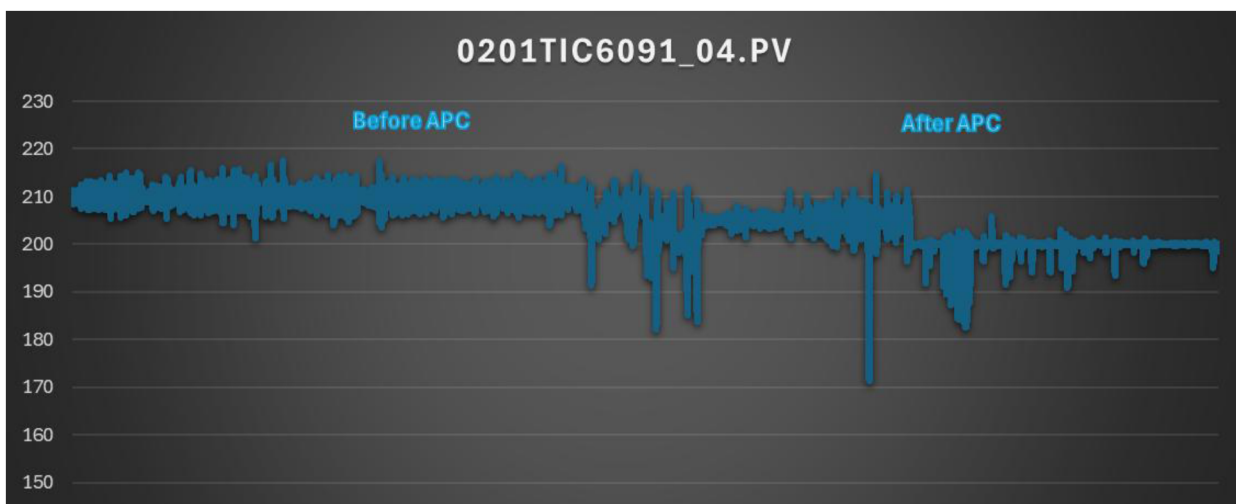
This section presents the performance trends of APC and its impact on the process parameters after; recorded post commissioning of APC solution.



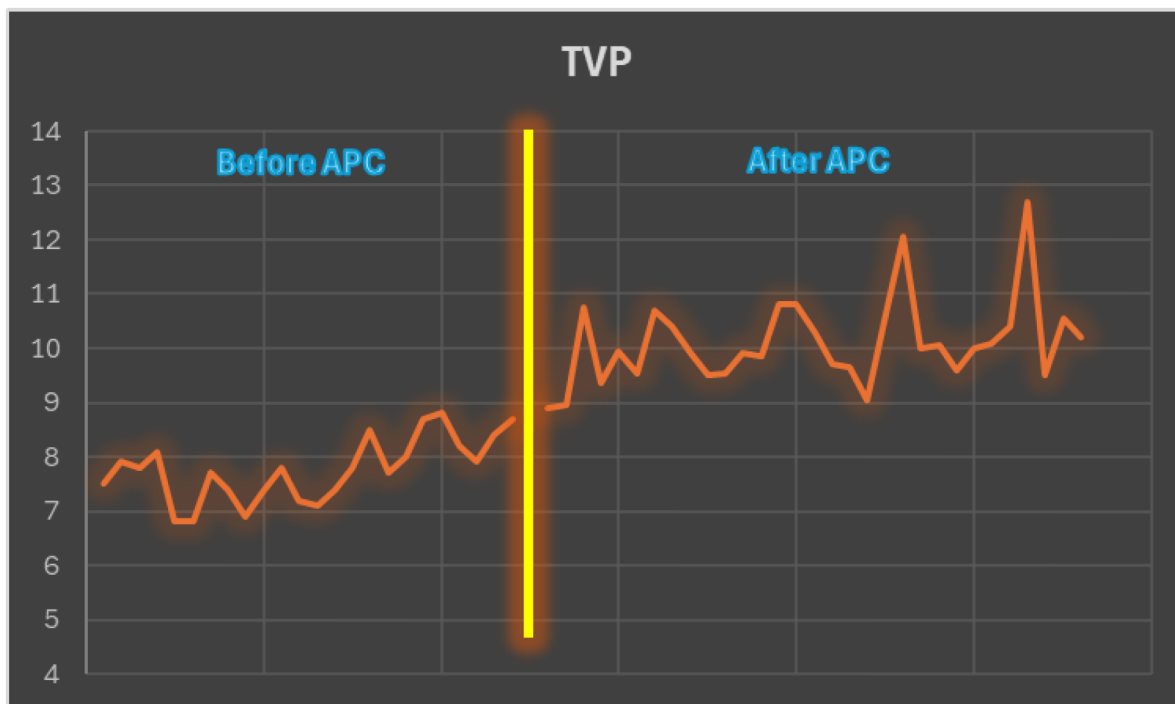
Trend-3—Train-3 Stabilizer Tray Temperature



Train-4—Train-4 Stabilizer Tray Temperature

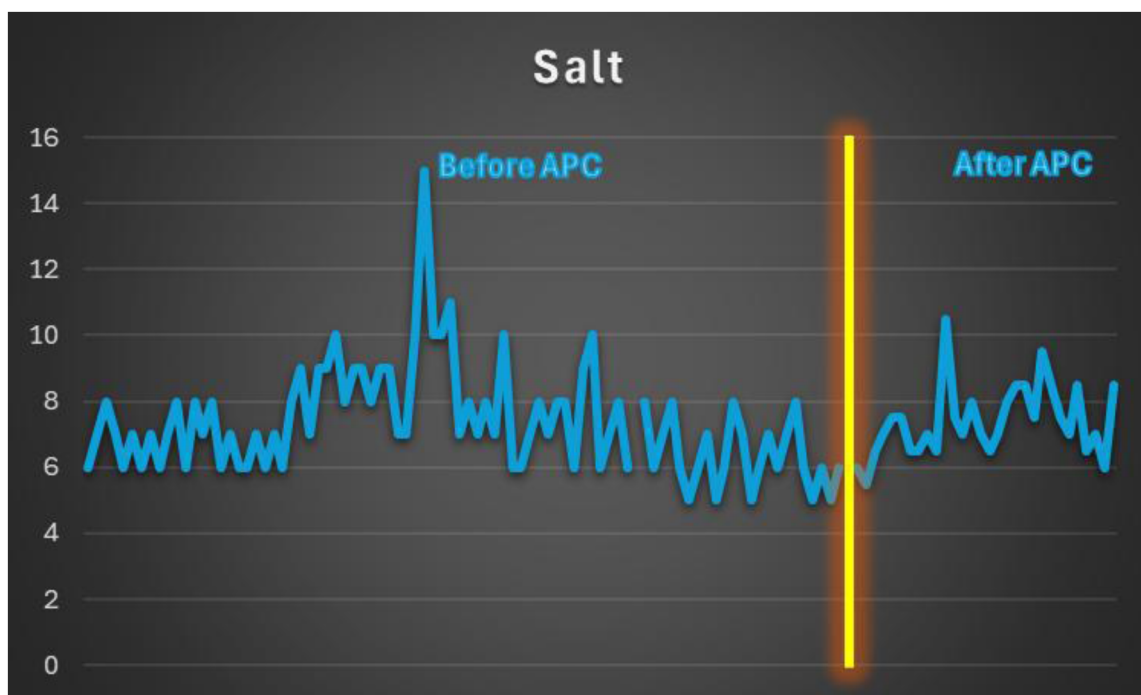


Trend-5—Hot Oil Heater Coil Outlet Temperature



Trend-6—TVP of Crude Oil

Trend-7—H₂S Content in Crude Oil



Trend-8—Salt Contents in Crude Oil

Conclusion

In conclusion, the deployment of Ai enabled APC not only achieved the target KPIs for production enhancement, energy optimization, and autonomous operations but also significantly improved operational stability under dynamic process conditions.

This paper introduces the pioneering application of AI-enabled APC in the upstream oil & gas industry. Unlike traditional APCs, this project uses modern day data analytics to predict plant performance under dynamic conditions and employs disturbance-based control strategies to stabilize processes, enhance production and optimize energy consumption in real-time. It has unlocked the new horizon of operational excellence in oil and gas upstream sector and serves as a key milestone towards achieving autonomous operations.

References

- *AGES-SP-04-013 – *Adnoc Group Projects And Engineering OT Cyber Security Specification*
- *AGES-SP-04-001 – *Adnoc Group Projects And Engineering – Process Control System Specification*
- J.M.Campbell Correlation to convert RVP to TVP.*
- Plant Process Data from Historian.*